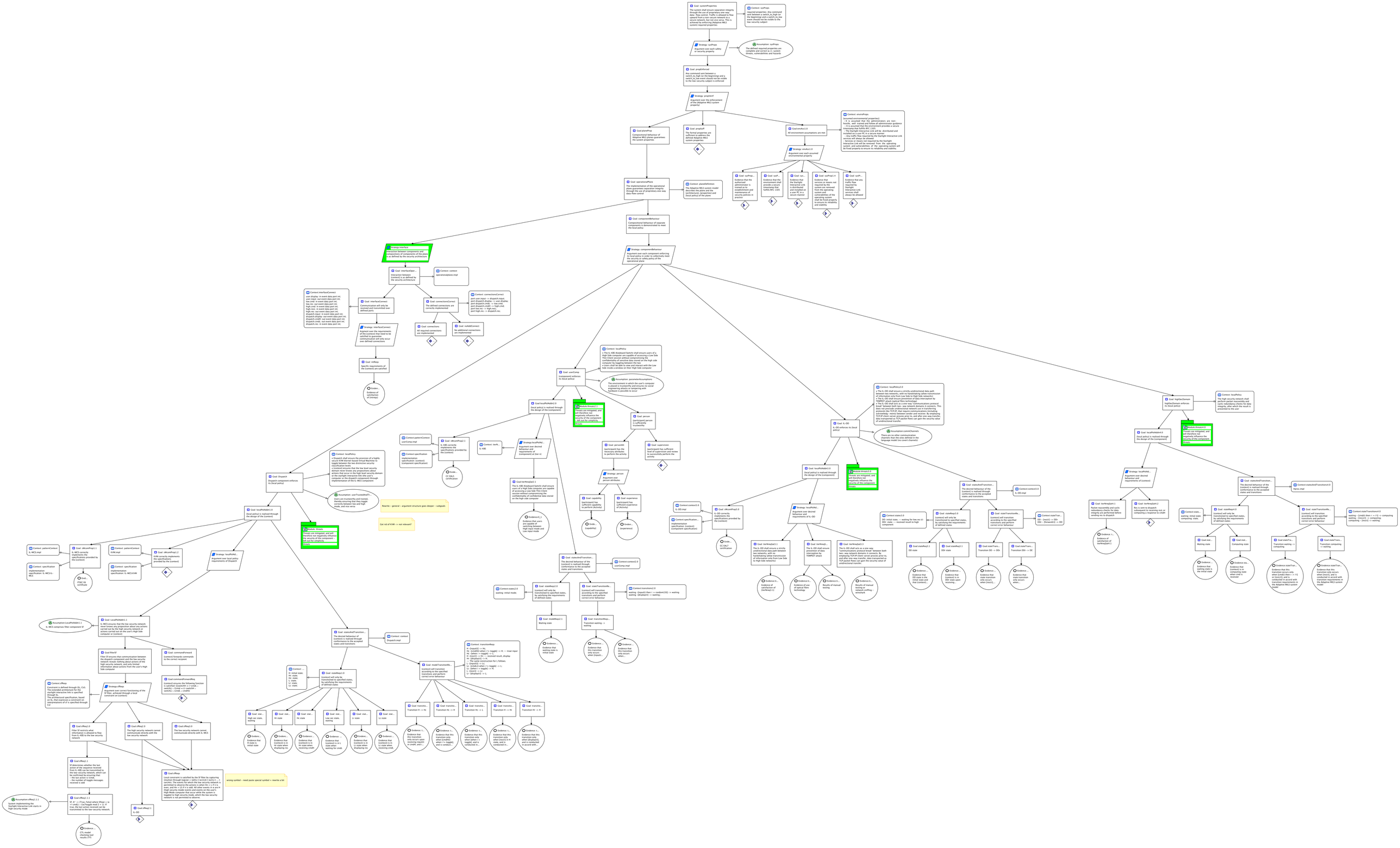
Goal: systemProperties
The system shall ensure separation integrity through the use of Adaptive MLS system properties. The system shall prevent the flow upward from a non-secure network to a secure network. Specifically, this is achieved by enforcing Adaptive MLS system required properties.
Context: sysProps
Assumption: sysProps: Any command sent between a switch_to_high (or the beginning) and a switch_to_low event should not be visible to the low security subject
Strategy: sysProps
Argument over each safety or security property
Assumption: sysProps
The defined required properties are complete and correct w.r.t. system threats, vulnerabilities and hazards
Goal: propsEnforced
Any command sent between a switch_to_high (or the beginning) and a switch_to_low event should not be visible to the low security subject is enforced
Strategy: progsreqf
Argument over the enforcement of Adaptive MLS system properties
Goal: planeProp
Compositional behaviour of Adaptive MLS planes guarantees the system properties
Goal: propsAll
The formal properties are sufficient to address the defined Adaptive MLS system properties
Goal: envAss1.0
All environment assumptions are met
Context: envAssProps
(Environmental properties):
- It is assumed that the administrators are non-hostile, well trained and follow all administrator guidance. The Sigrlight Interactive Link component provides a secure timesharing that fulfills RFC 1355
- The Sigrlight Interactive Link will be distributed and installed in a secure manner. The operating system and vulnerabilities of the operating system will be fixed properly to ensure its reliability and stability.
Strategy: envAss1.0
Argument over each assumed environmental property
Goal: sysProp...
Evidence that the administrator is provided with the installation and maintenance of security policies in process
Goal: sysP...
Evidence that the Sigrlight provide a secure timesharing that fulfills RFC 1355
Goal: sys...
Evidence that the Sigrlight Interactive Link is installed and installed as a user PC in a secure manner
Goal: sysProp1.4
Evidence that the operating system required by the Sigrlight Interactive Link from the operating system and vulnerabilities of the operating system shall be fixed properly to ensure its reliability and stability
Goal: sysP...
Evidence that only Sigrlight Interactive Link services shall always be allowed
Goal: operationalPlane
The implementation of the operational plane guarantees separation integrity through the use of proprietary one-way data-flow control
Context: planeDefinition
The Adaptive MLS system model describes the plane and the operational behaviour required (local policy) of the plane
Goal: componentBehaviour
Compositional behaviour of separate components is demonstrated to meet the local policy
Strategy: interface
Interaction between components and their environment is specified and is as defined by the security architecture
Strategy: componentBehaviour
Argument over each component enforcing its local policy in order to collectively meet the overall or safety policy of the operational plane
Goal: interfaceOper...
Interaction between components is specified by the security architecture
Context: context operationalplane.impl
Context: interfaceConnect
user.display: in event data port int;
user.input: out event data port int;
high.cmd: in event data port int;
low.rec: out event data port int;
high.cmd: in event data port int;
high.rec: out event data port int;
dispatch.input: in event data port int;
dispatch.output: out event data port int;
dispatch.cmd0: out event data port int;
dispatch.cmd1: out event data port int;
dispatch.rec: in event data port int;
Goal: interfaceConnect
Communication will only be allowed and transmitted over defined port
Goal: connectionsCorrect
The defined connections are correctly implemented
Context: connectionsCorrect
port user.input -> dispatch.input;
port dispatch.output -> user.display;
port dispatch.cmd0; -> low.cmd;
port dispatch.cmd0' -> high.cmd;
port low.rec -> dispatch.rec;
port high.rec -> dispatch.rec;
Strategy: interfaceConnect
Argument over the requirements of the (context) that need to be satisfied to guarantee communication is correctly occur over defined connections
Goal: connections
All required connections are implemented
Goal: noAddConnect
No additional connections are implemented
Goal: infReqs
Specific requirements of the (context) are satisfied
Evidence...
Evidence of satisfaction of (infreqs)
Goal: userComp
(component) enforces its (local policy)
Context: localPolicy
• The IL-4BS Wideband Switch shall ensure users of a High Side computer are capable of accessing a Low Side Thin Client session without compromising the confidentiality of data stored on the high side computer by toggling between the two
• Users shall be able to view and interact with the Low Side inside a window on their High Side computer
Assumption: parameterAssumptions
The environment in which the user's computer is placed is trustworthy and ensures no social engineering attacks or tampering with hardware is possible to occur
Goal: localPolicyAdd1.0
(local policy) is realized through the design of the (component)
Module threats1.0
Threats are mitigated, and are therefore not negatively influence the security of the component
Threats
Goal: person
(participant) person is sufficiently trustworthy
Goal: personAff
(participant) has the sufficient capability to perform the activity
Strategy: person
Argument over person attributes
Goal: capability
(participant) has sufficient capability to perform (Activity)
Evide...
(capability)
Goal: experience
(participant) has sufficient experience of (Activity)
Evidence...
(experience)
Goal: supervision
(participant) has sufficient supervision in place in order to successfully perform the activity
Context: patternContext
userComp.impl
Goal: IL-4BS
IL-4BS correctly implements the specifications provided by the (context)
Evide...
CC EALS certification
Context: specification
implementation specification: (context) (component specification)
Context: IL-4BS
Strategy: localPolicyAd...
Argument over desired behaviour and requirements of (component) or (for n)
Goal: toreRequest2.1
The IL-4BS Wideband Switch shall ensure users of a High Side computer are capable of accessing a Low Side Thin Client session without compromising the confidentiality of confidential data stored on the high side computer
Evidence1.1
Evidence that occurs are capable of switching between Low Side Thin Client and Low input mode
Goal: statesAndTransition...
The desired behaviour of the (context) is realized through the definition of the required states and transitions
Context:context1.0
userComp.impl
Context: localPolicy
• Dispatch shall ensure the provision of a highly secure communication channel that allows a toggle between the two distinctive security classification levels
• Dispatch shall ensure that the low level security domain never knows any propositions about actions that occur in the high level security domain
• Dispatch shall ensure that the user's computer or the dispatch component through implementation of the IL-MCS component
Goal: Dispatch
Dispatch (component enforces its (local policy)
Assumption: userTrustedAndTi...
Users are trustworthy and trained, and are able to toggle the Dispatch correctly between low and high mode, and vice versa
Rewrite - general - argument structure goes deeper - subgoals
Cat rid of KVM -> not relevant?
Goal: localPolicyAdd1.0
(local policy) is realized through the design of the (context)
Module: threats
Threats are mitigated, and are therefore not negatively influence the security of the component
Threats
Context: patternContext
IL-MCS.impl
Goal: IMcomProp1.1
IL-MCS correctly implements the specifications provided by the (context)
Context: specification
implementation specification: IL-MCS.IL-MCS
Evide...
FIPS 140 certificate
Context: patternContext
KVM.impl
Goal: IMcomProp1.2
KVM correctly implements the specifications provided by the (context)
Context: specification
implementation specification: IL-MCS.KVM
Strategy: localPolicyAd...
Argument over local policy requirements of Dispatch
Goal: localPolicyAdd1.1
IL-MCS ensures that the low security network cannot communicate directly with IL-MCS. Communication is carried out by the high security network or actions carried out on the user's High Side computer or (context)
Assumption: LocalPolicyAdd1.1
IL-MCS comprises filter component SF
Goal: FilterSF
Filter SF ensures that communication between the dispatch component and the low security network reveals nothing about actions of the high security domain and that no information about actions from the user's High Side computer
Goal: commandForward
(context) forwards commands to the correct recipient
Goal: commandForwardReq
(context) ensures the following function is satisfied: (CmdInH == i->cmd, ; CmdInL == i->cmd, ; CmdOutH == switch(i) -> cmd0, ; cmd0)
Context: sfReqs
Constraint is defined through (SL, CcI). The extended architecture for the starlight interactive link is specified through (SF). The architecural specification, based on SL, that expresses a constraint on implementation of (f) is specified through CcI
Strategy: sfReqs
Argument over correct functionning of the SF filter, achieved through a local constraint on (context)
Goal: sfReq1.0
Filter SF restricts what information can cross the low from IL-4BS to the low security network
Goal: sfReq2.0
The high security network cannot communicate directly with the low security network
Goal: sfReq3.0
The low security network cannot communicate directly with IL-MCS
Goal: sfReq1.1
Action SF performs on the last element of the sequence received from IL-4BS can be transmitted to the low security network. This can be confirmed by ensuring that:
- the last action is toreq0,
- the sequence of toggle messages received is odd
Goal: sfReq1.1.1
SF A' := (True, False) where (fAnal = (a -> cmd) -> (toc!toggle mod 2 = 1)). If true, the last action received can be transmitted to the low security network.
Assumption: sfReq1.1.1
System implementing the Starlight Interactive Link starts in high security mode
Evidence...
CTL model checking tool results (??)
Goal: sfReq1.2
IL-4BS
Goal: sfReq
Local constraint is satisfied by the SF filter by capturing the relevant events from the high side and low side (actns). The events for which the low security network is permitted to observe the actions is when H= i ∧ F = ∅ ∧ (i ∈ Ll ∨ i ∈ Lc). This means that the actions of the high security model events and events on the user's High Side computer that occur while the system is operating in high security mode, which the low security network is not permitted to observe.
wrong symbol - need paste special symbol + rewrite a bit
Goal: localPolicyAdd1.0
(local policy) is realized through the design of the (component)
Module threats1.0
Threats are mitigated, and are therefore not negatively influence the security of the component
Threats
Context: localPolicy1.0
• The IL-OD shall ensure a strictly unidirectional data path between two networks, with no handshaking (allow transmission of information only from Low Side to High Side network)
• The IL-OD shall prevent data interception by TEMPEST attack (optical fibre technology)
• The IL-OD shall act as a one-way "communications protocol breaker" between both lan- way network domains it connects. It does not preclude unidirectional network use in transferring protocols like TCP/IP, that require communications (including acknowledgments) between sender and receiver. For example, TCP/IP client-server process prior to, and after one-way transfer, data transmitted as TCP packet flows can gain the security value of unidirectional transfer.
Assumption:comsChannels
There are no other communication channels that are not specified in the language model (no cover channels)
Goal: IL-OD
IL-OD enforces its (local policy)
Goal: localPolicyAdd1.0
(local policy) is realized through the design of the (component)
Module threats1.0
Threats are mitigated, and are therefore not negatively influence the security of the component
Threats
Strategy: localPolicyAd...
Argument over desired behaviour and requirements of IL-OD
Goal: toreReq1.1
The IL-OD shall ensure a strictly unidirectional data path between two networks, with no handshaking (allow transmission of information only from Low Side to High Side network)
Evidence to satisfaction of (toreReq1.1)
Goal: toreReq2...
The IL-OD shall ensure data interception by TEMPEST attack
Evidence to...
Evidence of use of optical fibre technology
Goal: toreReq3.1
The IL-OD shall act as a one-way "communications protocol breaker" between both lan- way network domains it connects. By employing TCP/IP client-server process prior to, and after one-way transfer, data transmitted as TCP packet flows can gain the security value of unidirectional transfer.
Evidence to...
Results of manual testing or network sniffing ... wireshark
Goal: statesAndTransition...
The desired behaviour of the (context) is realized through definition of the required states and transitions
Context:context1.0
IL-OD.impl
Context:states3.0
DD: initial state. -- waiting for low.rec (!) ; DDr: state -- received result in high component
Goal: stateReq3.0
(context) will only be transitioned to specified states, by satisfying the requirements of defined states
Goal: stateReq1.1
DD state
Evidence...
Evidence that (context) is in DD state upon receiving...
Goal: stateReq1.2
DDr state
Evidence...
Evidence that (context) is in DDr state upon receiving...
Goal: stateTransitionRe...
(context) will transition according to the specified transitions and perform correct error behaviour
Context:stateTran...
DD -> (!rec!) -> DDr ; DDr -> (!forward!) -> DD
Goal: startTrans...
Transition DD -> DDr
Evidence...
Evidence that state transition only occurs when...
Goal: startTrans...
Transition DDr -> DC
Evidence...
Evidence that state transition only occurs when...
Goal: highSecDomain
highSecDomain enforces its (local policy)
Context: localPolicy
The high security network shall perform packet reassembly and cyclic redundancy checks for data integrity, after which the result is presented to the user
Goal: localPolicyAdd1.0
(local policy) is realized through the design of the (component)
Module threats1.0
Threats are mitigated, and are therefore not negatively influence the security of the component
Threats
Strategy: localPolicyAd...
Argument over desired behaviour and requirements of (context)
Goal: toreReqLat.1
Packet reassembly and cyclic redundancy checks for data integrity, after which before sending rec to dispatch
Evidence -...
Evidence of satisfaction of (toreReqLat.1)
Goal: toreReqLat.2
Rec to sent to dispatch subsequent to receiving req or computing a response to cmd
Context: state...
waiting: initial state; computing: state
Goal: stateReq1.1
(context) will only be transitioned to specified states, by satisfying the requirements of defined states
Goal: state...
Waiting state
Evidence...
Evidence that waiting state is the initial state
Goal: state...
Computing state
Evidence ob...
Evidence that (context) is in computing state upon receiving a request
Goal: statesAndTransition...
The desired behaviour of the (context) is realized through definition of the required states and transitions
Context:statesAndTransitions4.0
reprec.impl
Goal: stateTransitionRe...
(context) will transition according to the specified transitions and perform correct error behaviour
Context:stateTransitions4.0
waiting -> (indp) then r := rq1 -> computing ; waiting -> (rec!r) -> computing; computing -> (rec!r) -> waiting
Goal: state/Tra...
Transition waiting -> computing
Evidence stateTran...
Evidence that this transition occurs only when (indp) then r := rq1 or (rec!r), and is conducted in accord with transition requirements in the Adaptive MLS system model.
Goal: state/Tra...
Transition computing -> waiting
Evidence stateTran...
Evidence that this transition only occurs when (rec!r), and is conducted in accord with transition requirements in the Adaptive MLS system model.
Context: statesAndTransitions4.0
Context
M: initial state;
Hr: state;
Hc: state;
Ll: state;
Lr: state;
Lc: state;
Goal: stateReq1.0
(context) will only be transitioned to specified states, by satisfying the requirements of defined states
High sec state, waiting
Evidence... H state is initial state
Goal: sta...
Hr state
Evidence... (context) is in Hr state when "(toReq rq)"
Goal: sta...
Hc state
Evidence... (context) is in Hc state when "(toReq cmd)"
Low sec state, waiting
Evidence that (context) is in L state when waiting for cmd;
Goal: sta...
Lr state
Evidence... (context) is in Lr state when "(toReq rq)"
Goal: sta...
Lc state
Evidence... (context) is in Lc state when "(toReq cmd)"
Goal: modeTransitionRe...
(context) will transition according to the specified transitions and perform correct error behaviour
Context: transitionReqs
H: (input0) -> Hc;
Hc: (input0) -> (toggle) -> H; -- treat input
Hc: (when i -> toggle) -> L;
H: (input1) -> Hr; -- received result, display
Hr: (display0) -> H;
-- The same construction for L follows.
L: (input0) -> Lc;
Lc: (input0) -> (toggle) -> L;
Lc: (when i -> toggle) -> H;
L: (input1) -> Lr;
Lr: (display0) -> L;
Goal: transitio...
Transition H -> Hc
Evidence i... Evidence that transition only occurs upon receiving (input0), or (cmd0), and i...
Goal: transitio...
Transition Hc -> H
Evidence i... Evidence that this transition occurs when (input0) and (toggle) and is conducta...
Goal: transitio...
Transition Hc -> L
Evidence that this transition occurs when (when i -> toggle) and is conducted in...
Goal: transitio...
Transition H -> Hr
Evidence i... Evidence that this transition occurs when (input1) or (cmd1) and is conducted in H...
Goal: transitio...
Transition Hr -> H
Evidence i... Evidence that this transition occurs when (display0), and is conducted in accord with...